Department of Artificial Intelligence and Data Science (ADS)

Green University of Bangladesh

Semester: Spring 2026, B.Sc. in ADS

"DAILY LIFE AND PRODUCTIVITY TRACKER"

Group Project Report No: 01

Course Title: Introduction to Computational thinking and Programming

Course Code: ADS 100

Section: 261 D2

Students Details

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Lab Report Status

Marks: **Signature:**

Comments: **Date:**

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1. INTRODUCTION OF THE GROUP PROJECT:

"The Daily Life and Productivity Tracker" is a Python-based console application designed to help individuals organize their professional and personal lives. In a fast-paced academic and work environment, managing time and finances efficiently is critical. This project integrates three core pillars of productivity Task Management, Financial Tracking, and Goal Achievement into a single, easy-to-use interface. By quantifying productivity through a custom "Productivity Score," the system provides users with actionable insights into their daily performance.

2. Objective/Aim:

The objectives of this project are:

1. To provide a robust financial module for monitoring income, categorized expenses, and net savings in BDT.
2. To implement a progress-based goal-tracking system using visual progress bars.
3. To apply computational thinking principles such as decomposition (breaking down the app into menus) and abstraction (using data structures to represent real-world entities).

3. PROCEDURE:

The project was developed following the steps below:

1. **Requirement Analysis:** Identified the core needs of a student or professional (tasks, money, and long-term goals).
2. **Data Structure Design:** Utilized Python **Lists** to store records and **Dictionaries** to manage complex data attributes (e.g., amount, category, status).
3. **Modular Programming:** Divided the application into functional modules: task_manager, expense_tracker, and goal_tracker to ensure code reusability and clarity.
4. **Logic Implementation:** Developed loops for menu navigation and conditional logic for productivity score calculations.

5. **UI/UX Formatting:** Used helper functions like divider() and header() to create a clean, readable console interface.

4. Code Overview:

The code is built on a modular architecture where data is stored in global lists and manipulated through specific functions

Key Functional Components

• Task **Module:** Uses boolean logic (`True/False`) to toggle task completion and filters lists to show pending versus completed items.

• Finance **Module:** Performs real-time arithmetic to calculate total income, expenses, and current savings/deficit.

• Goal **Module:** Uses a visualization algorithm to render a 10-segment progress bar (█ and ░) based on percentage completion.

• Summary **Engine:** A unique algorithm that weights task completion (50%), financial health (30%), and goal milestones (20%) to generate a daily productivity score out of 100.

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5. TEST RESULT / OUTPUT:

Input:

- **Task: "Complete ADS100 Report"**
- **Income: 5000 BDT (Source: Allowance)**
- **Expense: 200 BDT (Category: Food)**
- **Goal: "Read 10 books" (Progress update: 2)**

OUTPUT:

```
==================================================
   YOUR DAILY SUMMARY
==================================================

Tasks:
--------------------------------------------------
  Total   : 1
  Done    : 0 (0%)
  Pending : 1

Finances:
--------------------------------------------------
  Income   : 5000.0 BDT
  Expenses : 200.0 BDT
  Savings  : 4800.0 BDT

Goals:
--------------------------------------------------
  Active    : 1
  Completed : 0

Productivity Score:
--------------------------------------------------
  [███░░░░░░░] 30/100
  Room to grow. You got this!
```

6. ANALYSIS AND DISCUSSION:

The system effectively demonstrates the power of Python for automation. During testing, the Financial Tracker correctly identified deficits when expenses exceeded income, providing a warning to the user. The Goal Tracker visualizes progress, which serves as a psychological motivator. The project successfully implements fundamental programming concepts:

- Iteration: Used for displaying lists and calculating totals.
- Validation: Basic checks for valid menu choices and task numbers.
- Data Persistence (Volatile): Currently, data is stored in RAM; while efficient for a single session, it resets upon exiting.

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7. Conclusion

The "Daily Life and Productivity Tracker" provides a streamlined solution for personal management. By integrating financial awareness with task accountability, it fulfills the requirements of the ADS 100 course by showcasing a functional application of Python programming and computational logic. It is a practical tool that can be immediately used for basic daily organization.

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8. FUTURE IMPROVEMENTS

- File I/O: Implementing JSON or CSV file handling to save user data permanently between sessions.
- GUI Transition: Developing a Graphical User Interface (GUI) using Tkinter for a more modern experience.
- Data Analytics: Adding Matplotlib to generate visual graphs of spending habits over time.
- Reminder System: Integrating time-based notifications for pending tasks.

9. REFERENCES

- Python Official Documentation: https://docs.python.org/3/
- Course Lectures: Intro to Computational Thinking (ADS 100).